

# Junaid Ahmed Mohammed

Senior Data Engineer

Indianapolis, IN | [mjunaidtechs@gmail.com](mailto:mjunaidtechs@gmail.com)

[linkedin.com/in/mjunaid56](https://linkedin.com/in/mjunaid56) | [github.com/Junaid-Techie](https://github.com/Junaid-Techie)

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## PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of experience designing and operating enterprise-scale data platforms on Azure, Snowflake, and Databricks. Built and governed high-throughput ETL/ELT pipelines and modern lakehouse architectures processing 12 TB+ of retail data across 625+ stores in 60+ countries, delivering 40% pipeline runtime reduction and 25% infrastructure cost savings. Deep expertise in PySpark, Delta Lake, Kafka, Airflow, dimensional modeling, and multi-source enterprise data integration. Experienced in enabling AI/ML workloads through vector search pipelines and LLM data preparation.

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## CORE COMPETENCIES & TECHNICAL STACK

**Data Engineering:** ETL/ELT, Data Warehousing, Lakehouse Architecture, Dimensional Modeling, SCD Type 1/2, Data Governance, Data Quality, Performance Optimization, CI/CD, Unit/Regression Testing, Source-to-Target Reconciliation, DataOps

**Cloud & Platforms:** Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Blob Storage, Snowflake (Zero-Copy Cloning, Time Travel), Azure DevOps, Azure Key Vault, Azure ML

**Data Processing:** Apache Spark, PySpark, Spark SQL, Delta Lake, Apache Kafka, Apache Airflow, Scala

**Programming & SQL:** Python, SQL, T-SQL, PL/SQL, Stored Procedures, Power BI, Scala

**Databases & Storage:** SQL Server, Oracle, PostgreSQL, Teradata

**AI/ML Enablement:** Vector Search, pgvector, OpenAI Embeddings API, LLM Fine-Tuning (QLoRA), Azure ML

**Tools & Collaboration:** Git, GitHub, Confluence, Azure DevOps

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## PROFESSIONAL EXPERIENCE

### NISUM TECHNOLOGIES, INC.

Nov 2018 – Dec 2023

Senior Data Engineer

Hyderabad, India

Led data engineering across 5 concurrent enterprise programs for Williams-Sonoma, Inc. (625+ stores, 60+ countries) and Macy's. Acted as primary technical decision-maker on pipeline architecture, data modeling standards, and Snowflake platform design, collaborating directly with WSI engineering managers and business stakeholders on requirements, timelines, and delivery.

- **Designed** scalable Azure-Snowflake lakehouse architectures supporting enterprise retail analytics, supply chain operations, customer delivery forecasting, and product management workloads across global business units spanning 625+ stores and 60+ countries.
- **Reduced** Databricks pipeline runtime by **40%** through PySpark optimization, partition pruning, broadcast joins, and Spark SQL tuning across 160+ ADF pipelines spanning a Bronze–Silver–Gold Delta Lake medallion architecture.
- **Co-architected** a Snowflake enterprise data warehouse and lakehouse consolidating **12 TB+** from Oracle, Teradata, Kafka, NetSuite ERP, and SFTP into ADLS Gen2, supporting daily batch and near-real-time ingestion across 6+ enterprise source systems.
- **Established** dimensional models, SCD Type 2 pipelines, clustering keys, and right-sized Snowflake virtual warehouses across the lakehouse, **cutting infrastructure costs by 25%**.
- **Owned the ECDD platform** (customer-facing delivery date model for Williams-Sonoma's global B2B and retail operations) — modeled carrier and routing dimensions with SCD Type 2 and applied SQL date calculation and exception-handling logic, **reducing delivery inaccuracies by 30%** and improving on-time fulfillment across 60+ countries.
- **Led** Oracle-to-Snowflake WMOS migration across warehouse, DBA, and data science teams; executed ADF orchestration, Databricks staging pipelines, Scala Snowflake loaders, and T-SQL stored procedures for metric computation.
- **Architected** a centralized PIM platform as single source of truth for enterprise product data, integrating Sterling OracleDB, SAP, and SFTP feeds across Orders, Products, Stores, and Vendor domains; drove cross-functional alignment between supply chain, merchandising, and engineering on data model requirements, delivering structured datasets to Power BI and retail analytics reporting layers.
- **Standardized** data quality, governance (SCD deduplication), and Git-based CI/CD deployments via Azure DevOps; enforced daily batch SLA compliance and secured pipeline credentials via Azure Key Vault.

- **Served as technical lead for 7 junior engineers** — conducted code reviews, defined pipeline engineering standards, and provided architecture guidance; coordinated feature dataset delivery into Azure ML workflows and maintained **100% on-time sprint delivery** across 5 concurrent projects and quarterly Agile releases.
- Client:** Macy's, Inc. | *Early-career data engineer supporting enterprise retail data warehouse.*
- **Monitored** 25+ ADF pipelines across daily batch windows; executed source-to-target reconciliation to detect data loss and duplication, debugged pipeline failures, and coordinated production fixes with senior engineers to maintain SLA compliance.
  - **Performed** unit and regression testing on pipeline transformations prior to production deployments; profiled 20+ operational tables in Oracle and SQL Server to certify data readiness and resolved historical reload issues.

**INDIANA UNIVERSITY INDIANAPOLIS**

Jan 2024 – Dec 2025

*Graduate Research Assistant, COMET Lab*

Indianapolis, IN

- Owned the data engineering workstream for applied ML and public health research, designing and operating scalable data pipelines and orchestration infrastructure independently.*
- **Architected** an end-to-end streaming ingestion platform using Kafka and PySpark Structured Streaming, validating **50K+ rows per batch cycle** against revenue, cost, date, and categorical business rules before loading into PostgreSQL.
  - **Orchestrated** automated model training via Airflow DAG with threshold-triggered deduplication and distributed Spark-submit jobs for revenue and units-sold forecasting; containerized the full stack with Docker Compose.
  - **Constructed** LLM training data pipelines and evaluation infrastructure for QLoRA fine-tuning of LLaMA 3.2 and Qwen 2.5; built structured prompt datasets and systematized correctness harnesses to validate domain-specific Python code generation.
  - **Delivered** multi-source data ingestion pipelines for NHANES and IPUMS public health datasets; standardized file formats using PySpark, enforced weighted aggregation logic, and streamlined report-ready data delivery for research and policy audiences.

**PERSONAL PROJECTS**

**JOB AI MATCHER** | *AI-Powered Semantic Job Intelligence Platform*

*In Development | Beta*

*Stack: Python, FastAPI, PostgreSQL, pgvector, OpenAI Embeddings API*

- **Built and Deployed** a cloud-hosted semantic job matching platform using FastAPI and PostgreSQL; engineered a vector search layer with pgvector and OpenAI embeddings to rank resumes against live job descriptions via similarity search.
- **Integrated** live job ingestion from multiple external APIs with an end-to-end pipeline covering semantic matching, ranking, and tailored cover letter generation via GPT-4o-mini.

**EDUCATION**

**INDIANA UNIVERSITY**

Indianapolis, Indiana

M.S. in Applied Data Science | GPA: 3.97 / 4.0

Aug 2023 – Dec 2025

*Coursework: Statistical Learning, Cloud Computing, NLP, Computer Vision, Database Design*

**CERTIFICATIONS**

Snowflake SnowPro Core | Azure Data Engineer Associate (DP-203) | Generative AI 101, Indiana University